

NLP_Practical_01

April 7, 2026

0.0.1 Import Libraries

```
[15]: import nltk
```

0.0.2 Download required resources (run once)

```
[16]: nltk.download('punkt')
nltk.download('wordnet')
nltk.download('omw-1.4')

from nltk.tokenize import word_tokenize, wordpunct_tokenize,
↳TreebankWordTokenizer, TweetTokenizer, MWETokenizer
from nltk.stem import PorterStemmer, SnowballStemmer
from nltk.stem import WordNetLemmatizer
```

```
[nltk_data] Downloading package punkt to
[nltk_data] C:\Users\NIKKI\AppData\Roaming\nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data] C:\Users\NIKKI\AppData\Roaming\nltk_data...
[nltk_data] Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to
[nltk_data] C:\Users\NIKKI\AppData\Roaming\nltk_data...
[nltk_data] Package omw-1.4 is already up-to-date!
```

0.0.3 Sample Text

```
[4]: text = "Hello! My name is Nikki. I'm learning NLP with NLTK. It's fun :) #AI
↳#MachineLearning"

print("Original Text:\n", text)
```

Original Text:

```
Hello! My name is Nikki. I'm learning NLP with NLTK. It's fun :) #AI
#MachineLearning
```

0.0.4 Tokenization

1. Whitespace Tokenization (using split)

```
[5]: whitespace_tokens = text.split()
print("\nWhitespace Tokenization:\n", whitespace_tokens)
```

Whitespace Tokenization:

```
['Hello!', 'My', 'name', 'is', 'Nikki.', 'I'm', 'learning', 'NLP', 'with',
'NLTK.', 'It's', 'fun', ':)', '#AI', '#MachineLearning']
```

2. Punctuation-based Tokenization

```
[6]: punct_tokens = wordpunct_tokenize(text)
print("\nPunctuation-based Tokenization:\n", punct_tokens)
```

Punctuation-based Tokenization:

```
['Hello', '!', 'My', 'name', 'is', 'Nikki', '.', 'I', '"', 'm', 'learning',
'NLP', 'with', 'NLTK', '.', 'It', '"', 's', 'fun', ':)', '#', 'AI', '#',
'MachineLearning']
```

3. Treebank Tokenization

```
[7]: treebank_tokenizer = TreebankWordTokenizer()
treebank_tokens = treebank_tokenizer.tokenize(text)
print("\nTreebank Tokenization:\n", treebank_tokens)
```

Treebank Tokenization:

```
['Hello', '!', 'My', 'name', 'is', 'Nikki.', 'I', '"m', 'learning', 'NLP',
'with', 'NLTK.', 'It', '"s', 'fun', ':', ')', '#', 'AI', '#', 'MachineLearning']
```

4. Tweet Tokenization

```
[8]: tweet_tokenizer = TweetTokenizer()
tweet_tokens = tweet_tokenizer.tokenize(text)
print("\nTweet Tokenization:\n", tweet_tokens)
```

Tweet Tokenization:

```
['Hello', '!', 'My', 'name', 'is', 'Nikki', '.', 'I'm', 'learning', 'NLP',
'with', 'NLTK', '.', 'It's', 'fun', ':)', '#AI', '#MachineLearning']
```

5. MWE Tokenization (Multi-Word Expression)

```
[9]: mwe_tokenizer = MWETokenizer([('Machine', 'Learning'), ('New', 'York')])
mwe_tokens = mwe_tokenizer.tokenize(text.split())
print("\nMWE Tokenization:\n", mwe_tokens)
```

MWE Tokenization:

```
['Hello!', 'My', 'name', 'is', 'Nikki.', 'I'm', 'learning', 'NLP', 'with',
'NLTK.', 'It's', 'fun', ':)', '#AI', '#MachineLearning']
```

0.0.5 2. Stemming

```
[12]: words = word_tokenize(text)

# Porter Stemmer
porter = PorterStemmer()
porter_stems = [porter.stem(word) for word in words]
print("\nPorter Stemming:\n", porter_stems)

# Snowball Stemmer
snowball = SnowballStemmer("english")
snowball_stems = [snowball.stem(word) for word in words]
print("\nSnowball Stemming:\n", snowball_stems)
```

Porter Stemming:

```
['hello', '!', 'my', 'name', 'is', 'nikki', '.', 'i', 'm', 'learn', 'nlp',
'with', 'nltk', '.', 'it', 's', 'fun', ':', ')', '#', 'ai', '#',
'machinelearn']
```

Snowball Stemming:

```
['hello', '!', 'my', 'name', 'is', 'nikki', '.', 'i', 'm', 'learn', 'nlp',
'with', 'nltk', '.', 'it', 's', 'fun', ':', ')', '#', 'ai', '#',
'machinelearn']
```

0.0.6 3. Lemmatization

```
[14]: lemmatizer = WordNetLemmatizer()

lemmatized_words = [lemmatizer.lemmatize(word) for word in words]
print("\nLemmatization:\n", lemmatized_words)
```

Lemmatization:

```
['Hello', '!', 'My', 'name', 'is', 'Nikki', '.', 'I', 'm', 'learning', 'NLP',
'with', 'NLTK', '.', 'It', 's', 'fun', ':', ')', '#', 'AI', '#',
'MachineLearning']
```

```
[ ]:
```